

	1	2	3	4	5	6	7	8
A	1. Connector & input protection Field connector, 2-stage bidirectional TVS surge protection, chassis isolation File: HMT500(260313A)_connector.kicad_sch		2. Power eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V File: HMT500(260313A)_power.kicad_sch		3. MCU STM32G0B1CCT3, SWD, reset, internal temperature sensor File: HMT500(260313A)_mcu.kicad_sch		4. Measurement Capacitive humidity sensor (PCAP04) and Pt1000 2-wire (ADS1220) File: HMT500(260313A)_measurement.kicad_sch	
B								
C	5. Analog outputs 2x DAC8760 V/I output with TPS26611 miswiring protection File: HMT500(260313A)_analog_out.kicad_sch		6. RS-485 THVD2410 +/-70V fault-protected transceiver File: HMT500(260313A)_rs485.kicad_sch		7. Aux TPS26611 +Vs clamp, supply monitor File: HMT500(260313A)_aux.kicad_sch			
D	DOTECH HMT500(260313) – Oil Moisture Transmitter – Schematic v0.9 – Project No. HMT500(260313A) File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A) Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2410 -> J1 Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-28 V DC. Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact. Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels. History v0.3-v0.8 (MCU G0B1, JST harness, DAC8760/TPS26611 datasheet pinouts, RTN isolation, gated DAC SCLK ...): docs/decision-log.md #9-#26. v0.9: 3-pass review (docs/hw/final-review-260313A.md): OVP 32.6 V (R5 36.5k) + TPS26611 +Vs follower clamp ~26.4 V; D2 -> TVS3301; eFuse IN C2; buck CIN; RS485_DE 10k pull-down (in-house bootloader, dec. #31); THVD2410; X7R only; R2 HV 1206; SGOOD -> PA0/PA4; CS_CDC -> PA15; VIN_SENSE PB0; LATCH/CS pull-ups; OPA197 V+ = DAC AVDD; R31/R41 100k; SWD no-connect flags removed; sheet Aux: +Vs clamp, VIN monitor (DAC boost / HSE / sensor ESD: no room, dec. #32). Footprints: KiCad 7.0.11 library + project library HMT500_260313A (LMR36006 RNX0012A, GDT Bourns 2035-xx-SM, TVS3301 pad floating). 표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울임 글자 (+3V3, VIN_P) 신호 네트 = 테두리 있는 라벨 (SPL_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음 라벨 모양 = 신호 방향: 뾰족한 쪽이 밖 = 이 사이트에서 내보냄(출력), 안 = 받음(입력), 양쪽 = 양방향, 네모 = 아날로그-수동 글자 겹침: 생성기가 부품 몸체·핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리							
E								
F	1	2	3	4	5	6	7	8

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /

File: HMT500(260313A).kicad_sch

Title: HMT500(260313A) Top

Size: A3

Date: 2026-09-30

Rev: 0.9

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 1/8

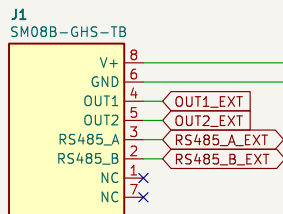
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)		
Design notes: docs/hw/circuit-design.md		
Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter		
DOTECH Co., Ltd.		
Sheet: /		
File: HMT500(260313A).kicad_sch		
Title: HMT500(260313A) Top		
Size: A3	Date: 2026-09-30	Rev: 0.9
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/8

FIELD CONNECTOR

현장 커넥터

J1 JST GH 8P -> W-2 -> M12 8P

EE364 pinout, pin n = pin n



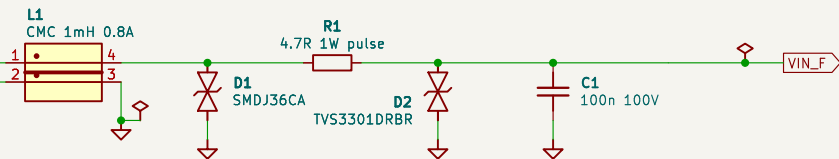
Pins 1, 7: not connected

GH mounting pads (MP): PCB only

INPUT SURGE / REVERSE-POLARITY PROTECTION

입력 서지·역극성 보호

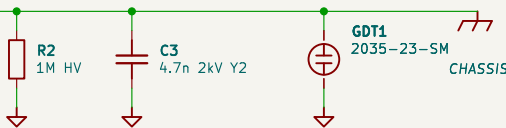
TVS bidirectional: -30 V miswiring must not conduct



SMDJ 3 kW (1st) -> R1 -> TVS3301 flat clamp (2nd) -> eFuse on sheet Power

CIRCUIT GND <--> CHASSIS (floating)

회로 GND-외함 분리



PCB-housing creepage >= 2 mm

GDT conducts only on line-to-ground surge

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
J1	SM08B-GHS-TB	현장 커넥터 연결	JST GH 1.25 mm 8P 옆 삽입, 하네스 W-2 -> M12 8P, 핀 n = M12 핀 n
L1	CMC 1mH 0.8A	전원선 공통모드 노이즈 차단	Würth WE-SL2 744222, 2 × 1 mH, 0.8 A, 9.2 × 6 × 5 mm — 전도 방출·내성 대책
D1	SMDJ36CA	입력 서지 1단 흡수	TVS 양방향 3000 W, 36 V — -30 V 오결선에 도통 안 함
R1	4.7R 1W pulse	1단·2단 서지 분담	MELF 1 W 펄스 내량 저항
D2	TVS3301DRBR	입력 서지 2단 클램프	TI 평탄 클램프 ±33 V, 42.5 V @ 27 A — 음(-) 서지에서 eFuse IN-OUT을 -70 V/10 ms 이내로 (전원 ≤ 28 V, 여유 작용 -> 물딩 전 서지 시험). DC 입력 33 V 초과 금지 (v0.9, 이전 SMBJ33CA)
GDT1	2035-23-SM	회로 GND-외함 서지 방전	Bourns 2전극 SMD GDT 230 V, Ø5 × 4.4 mm — 산-대지 서지 때만 도통
C3, R2	4.7n Y2 / 1M HV	GND-외함 고주파 결합	Y2 안전 콘덴서 + 1 MΩ 고전압 침 저항 1206 (사용 전압 ≥ 1 kV — GDT 방전 전 임펄스를 견딤, 물딩 안)
J5	CHASSIS wire	PCB-바디 접지 선	AWG 28 선 약 25 mm -> M2 링 단자 -> PCB 홀더 축 나사(금속 바디 탭). 조립 시뮬레이션 결과 스프링 접점 대체

* 표시 값은 데이터시트로 확인

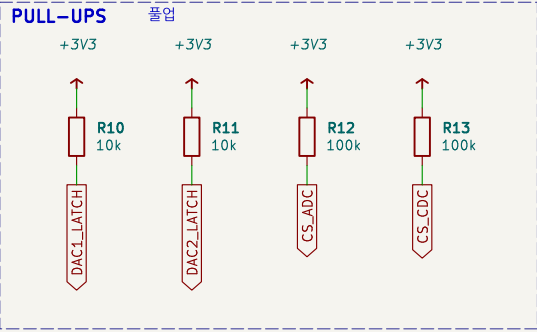
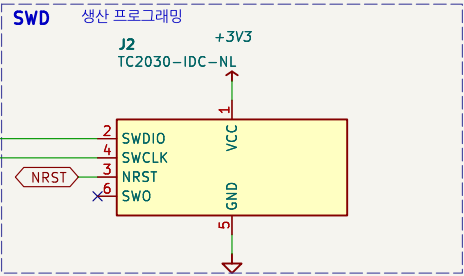
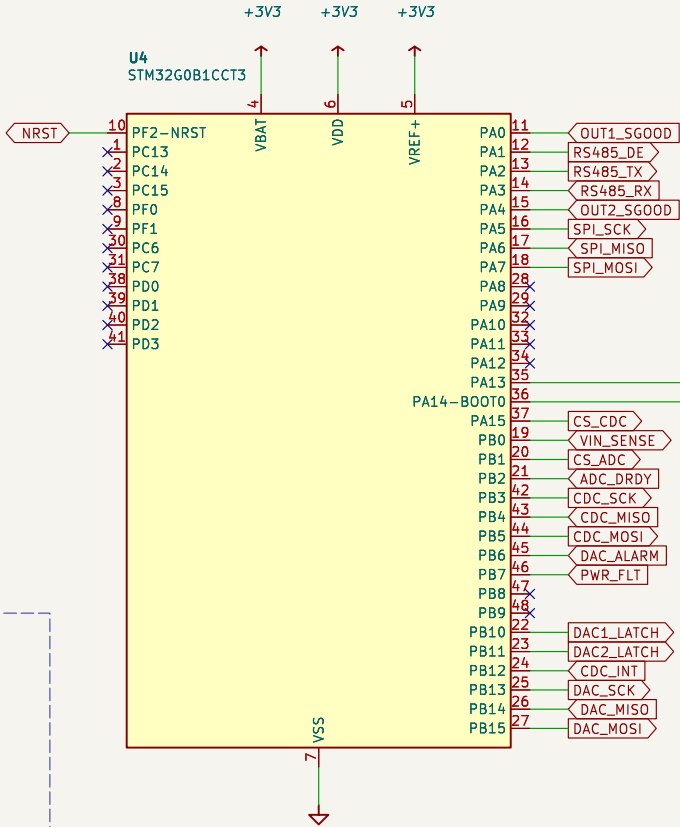
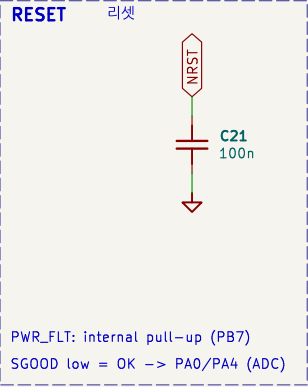
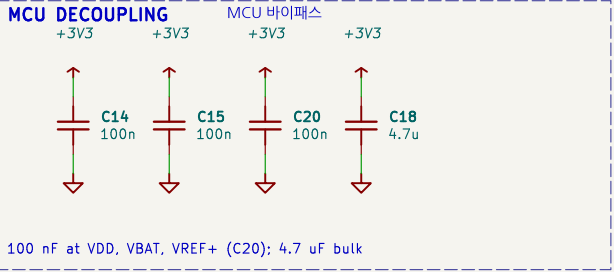
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /1. Connector & input protection/
File: HMT500(260313A)_connector.kicad_sch

Title: HMT500(260313A) Connector & input protection

Size: A3	Date: 2026-09-30	Rev: 0.9
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/8



MCU NOTES (STM32G0B1) MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1, IWDG_SW=0, RDP 1

-> in-house RS-485 bootloader 0x08000000 (dec. #31).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6): T3 = -40..125 C.

VIN_SENSE (PB0 ADC_IN8) = VIN_P / 11.

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U4	STM32G0B1CCT3	제어-통신-보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-125 °C, LQFP48, 내부 온도센서 (PCB 온도). SPI1 = ADS1220, SPI2 = DAC (모드 3, SCLK 게이트), SPI3 = PCAP04
J2	TC2030-IDC-NL	생산 매 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유
R10, R11	10k	DAC LATCH 풀업	리셋 중 LATCH High -> SCLK 게이트 닫힘, DAC 오입력 방지 (v0.9)
R12, R13	100k	CS 풀업	리셋 중 ADS1220-PCAP04 선택 해제 (디지털 임력 부동 금지)

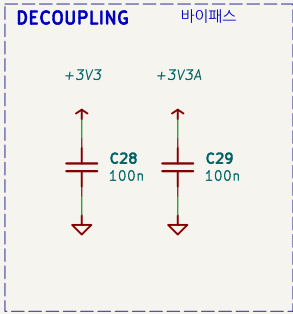
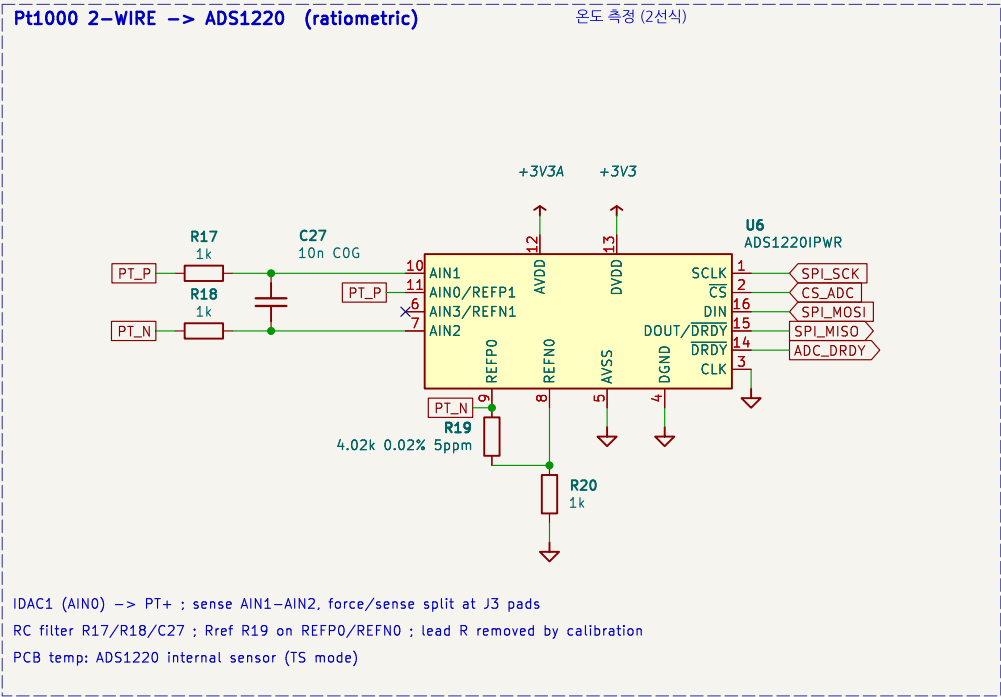
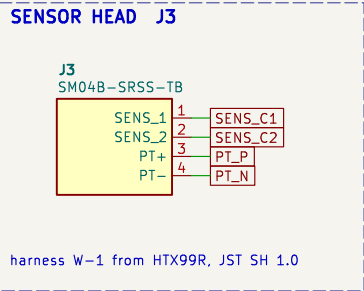
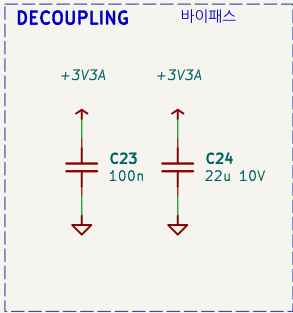
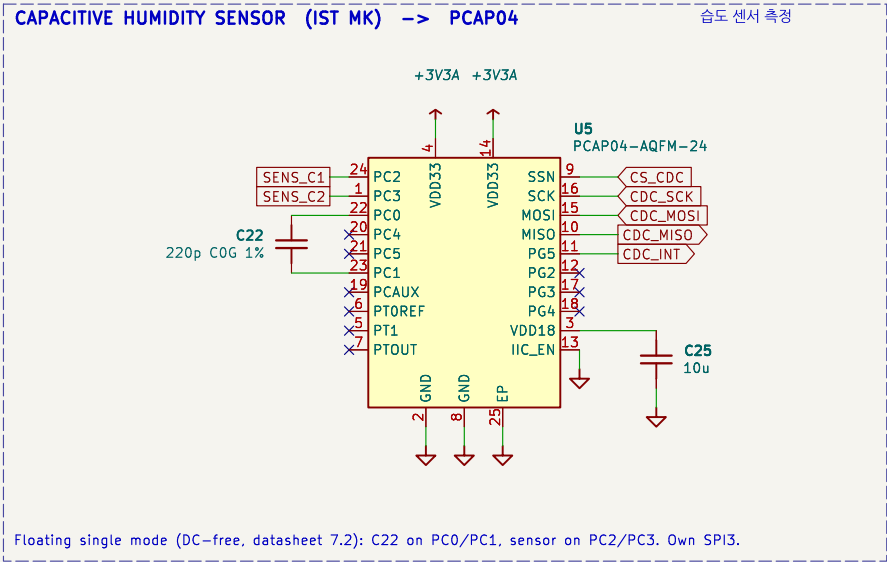
* 표시 값은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

Sheet: /3. MCU/
File: HMT500(260313A)_mcu.kicad_sch

Title: HMT500(260313A) MCU

Size: A3	Date: 2026-09-30	Rev: 0.9
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/8



주요 부품 — 역할·특징·사양			
부품번호	부품	역할	특징·사양
J3	SM04B-SRSS-TB	센서 하네스 W-1 연결	JST SH 1.0 mm 4P 옆 삽입, 1-2 = MK33, 3-4 = Pt1000
U5	PCAP04-AQFM-24	습도 센서 정전용량 측정	정전용량-디지털 변환, 기준 C와 비율 측정(방전 시간), DC 없는 구동, 내부 DSP, 전용 SPI3 (모드 1)
C25, C24	10u / 22u	PCAP04 전원 버퍼	VDD18 ≥ 4.7 uF, VDD33 ≥ 10 uF (데이터시트 표 2) — DC 바이어스 후에도 충족 (v0.9)
C22	220p COG 1%	PCAP04 기준 용량	MK33-W mini (200 pF)와 비슷한 값, COG, MK33-W (300 pF)면 270-330 pF (센서 품번 확정 후)
U6	ADS1220IPWR	Pt1000 온도 측정	24비트 ADC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도), 설정 한계: IDAC ≤ 250 uA, PGA ≤ 2 (SPI 모드 1)
R19	4.02k 0.02% 5ppm	Pt1000 비율 측정 기준저항	IDAC 전류 오차 상쇄, 온도계수 5 ppm/°C
R17, R18, C27	1k / 10n COG	입력 RC 필터	차동 노이즈 제거 (COG, 데이터시트 9.2)
* 표시 값은 데이터시트로 확인			

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

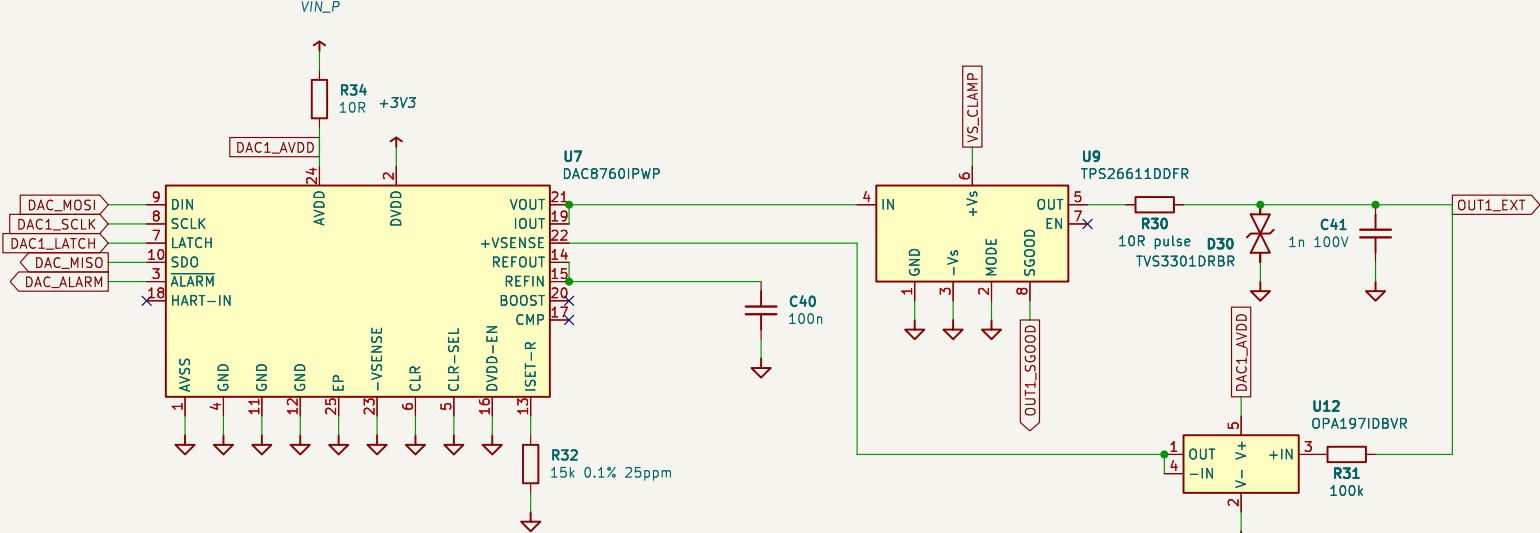
Sheet: /4. Measurement/
File: HMT500(260313A)_measurement.kicad_sch

Title: HMT500(260313A) Measurement

Size: A3	Date: 2026-09-30	Rev: 0.9
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/8

ANALOG OUTPUT CH1 - V/I selectable, miswiring protected -> connector pin 4

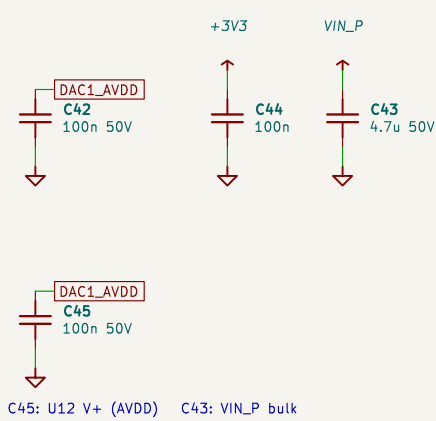
아날로그 출력 1



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U12 OPA197 follower of terminal OUT1_EXT (after R30) via R31:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R32 15k 0.1%. SGOOD low = OK.

CH1 DECOUPLING

채널 1 바이패스



DAC SELECTION (decision #10)

DAC 선정

U7/U8 fitted: DAC8760 (16-bit, I + V output).

Drop-in alternates (same pinout, no layout change):

DAC7760 12-bit, I + V (cost-down, V/I kept)

DAC8750 16-bit, I only 4-20/0-20/0-24 mA (EE364-equiv.)

DAC7750 12-bit, I only (-5 uA/step vs EE364 2 uA)

Current-only (x750): VOUT/+VSENSE/-VSENSE absent -

check pin handling before fitting (VERIFY).

DAC SPI - SPI2, SCLK gated per DAC

DAC 전용 SPI

DAC8760 rev D: daisy chain removed; multiple devices

need gated SCLK (datasheet 8.5.1.5, Fig. 8-9).

SPI2 mode 3 (CPOL 1, CPHA 1): SCK idles high, so the

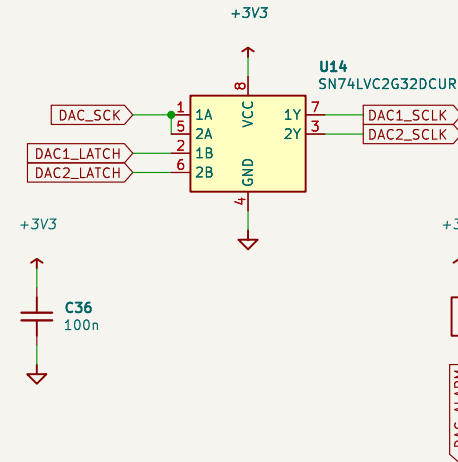
OR gate gives no extra edge at LATCH low/high.

DACn_SCLK = DAC_SCLK OR DACn_LATCH (PB10 / PB11).

DIN and SDO shared (SDO 3-state after LATCH).

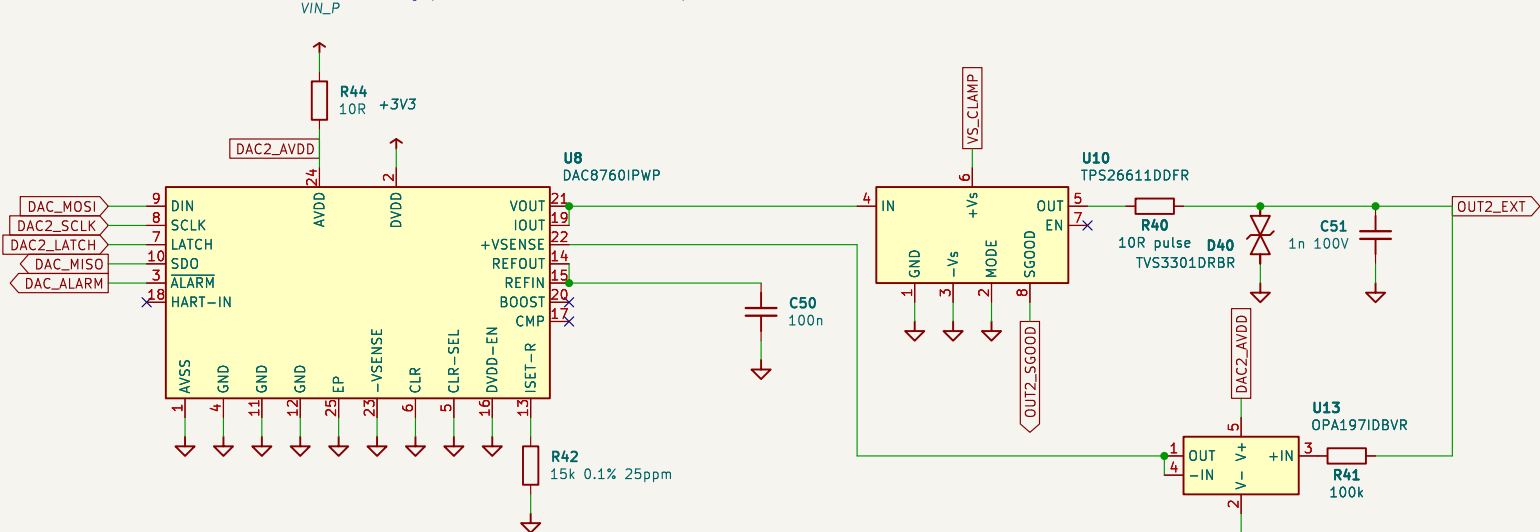
ALARM: open drain, wired-OR, R33 10k pull-up:

which channel -> read status register.



ANALOG OUTPUT CH2 - V/I selectable, miswiring protected -> connector pin 5

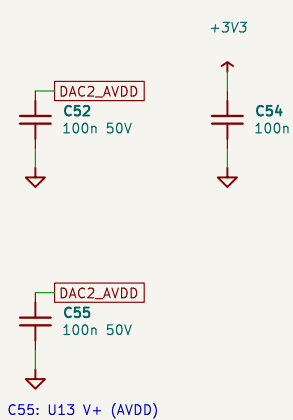
아날로그 출력 2



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U13 OPA197 follower of terminal OUT2_EXT (after R40) via R41:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R42 15k 0.1%. SGOOD low = OK.

CH2 DECOUPLING

채널 2 바이패스



주요 부품 - 역할·특징·사양

부품번호	부품	역할	특징·사양
U7, U8	DAC8760IPWP	아날로그 출력 1·2	16비트, 4-20 / 0-20 / 0-24 mA · 0-5 / 0-10 V 선택, 내부 기준, 고정 알람 — 대체 DAC7760/8750/7750
U9, U10	TPS26611DDFR	출력 오걸선 보호	±50 V, 단자 > +Vs 또는 < -0.2 V면 5 us 차단·자동 복귀, 전류 제한 32 mA, RON 7.5 Ω, SGOOD Low = 정상. +Vs = VS_CLAMP (27 V 제너, 시트 Aux)
R30, R40	10R pulse (2512)	서지 전류 제한	출력 보호기와 TVS 사이 직렬
D30, D40	TVS3301DRBR	출력 서지 클램프	TI 평탄 클램프 33 V 양방향, 40 V @ 27 A — TPS26611 ±55 V 보호 (데이터시트 권장)
C41, C51	1n 100V	출력 고주파 필터	케이블로 들어오는 RF 억제
U12, U13	OPA197IDBVR	+VSENSE 버퍼 (이득 1)	36 V, 입출력 레일투레일 — 단자 전압을 +VSENSE로, 전류 모드 누설 0. V+ = DAC AVDD
R31, R41	100k	버퍼 입력 보호	R30-R40 뒤 단자 전압을 OPA197 + 입력으로 (서지 전류 제한, ±30 V 오버션 9 mW, v0.9)
R32, R42	15k 0.1% 25ppm	외부 ISET-R (전류 기준)	내부 저항보다 온도 드리프트 작음 (REXT 비트 설정)
R34, R44	10R	DAC AVDD 직렬	전원 상승 속도 제한 (데이터시트 10장)
U14	SN74LVC2G32DCUR	DAC별 SCLK 게이트	OR 2회로: DACn_SCLK = SCK OR LATCHn (SPI2 모드 3)
R33	10k	ALARM 플립	오픈 드레인 ALARM 두 개를 한 선으로 (데이터시트: 외부 10 kΩ)

* 표시 값은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter

DOTTECH Co., Ltd.

Sheet: /5. Analog outputs/

File: HMT500(260313A)_analog_out.kicad_sch

Title: HMT500(260313A) Analog outputs

Size: A3

Date: 2026-09-30

Rev: 0.9

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 6/8

A

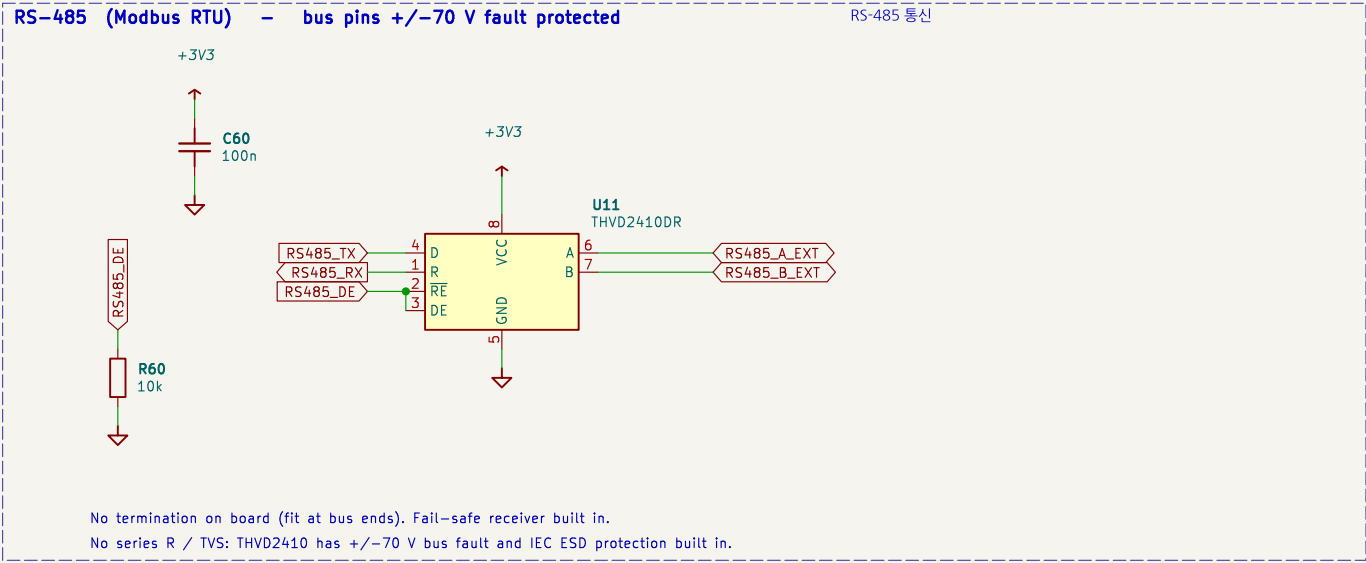
B

C

D

E

F



주요 부품 — 역할·특징·사양

부품번호	부품	역할	특징·사양
U11	THVD2410DR	RS-485 (Modbus RTU) 통신	3.3 V 반이중 500 kbps (느린 예지 → EMC·반사 유리, v0.9), 버스 핀 ±70 V 고장 보호, IEC ESD 보호 내장, 버스 개방 시 안전 수신
R60	10k	DE 풀다운	MCU 라셋·부팅·부트로더 점프 중 송신기 OFF·수신 ON (자체 부트로더 필수, 결정 #31)
C60	100n	전원 바이패스	U11 VCC

* 표시 값은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

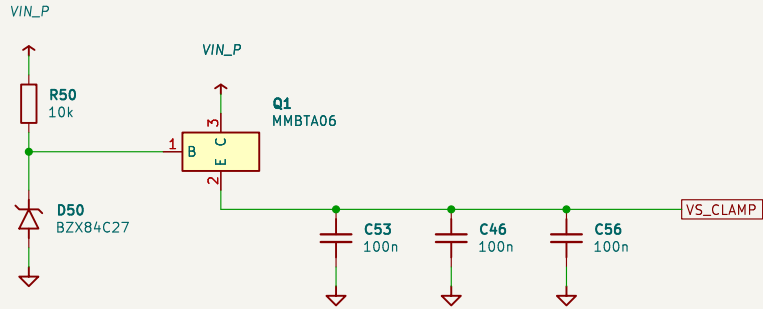
Sheet: /6. RS-485/
File: HMT500(260313A)_rs485.kicad_sch

Title: HMT500(260313A) RS-485

Size: A3	Date: 2026-09-30	Rev: 0.9
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 7/8

OUTPUT PROTECTOR SUPPLY CLAMP

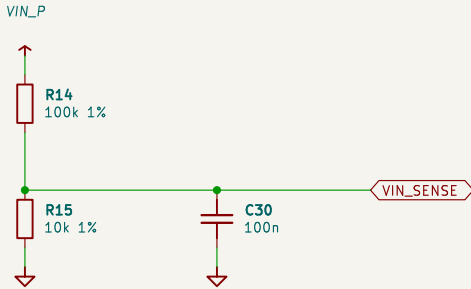
출력 보호기 전원 클램프



+Vs abs max 32 V; VIN_P up to 34.2 V (OVP max). Q1 follower: VS = min(VIN_P - 1, ~26.4 V).
C46 at U9 pin 6, C56 at U10 pin 6.

SUPPLY MONITOR

전원 전압 감시



VIN_SENSE = VIN_P / 11 -> MCU PB0 (ADC_IN8)

OPTIONS NOT FITTED ON PCB

PCB에 넣지 않은 선택 사항

DAC8760 IOUT boost (SBAS528D 8.4.2): no room next to U7/U8 without moving ISET-R / REF / AVDD parts.

-> worst case 0.73 W per DAC (28 V, 24 mA short): thermal test on a potted prototype;

if too hot: limit spec (supply ≤ 24 V or load ≥ 250 ohm) or add boost in next PCB rev.

HSE crystal: no room next to PF0/PF1 -> factory HSI trim + firmware HSI calibration

from Modbus frame timing (USART auto-baud / timer capture).

Sensor-line ESD array: no room next to J3 -> probe ESD test (IEC 61000-4-2) decides;

if needed: protect at the HTX99R side or in the next PCB rev.

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
Q1, R50, D50	MMBTA06 / 10k / BZX84C27	TPS26611 +Vs 클램프	이미터 풀로류: VS = VIN_P - 약 1 V, 최대 약 26.4 V (VIN_P 최대 34.2 V) — +Vs 절대 최대 32 V 보호, 낮은 전원에서 강하 작용. SOT-23 (Q1: 1 B, 2 E, 3 C / D50: 1 A, 3 K)
C53, C46, C56	100n	+Vs 바이패스	C46 → U9 6번, C56 → U10 6번 핀 옆 (TPS26611 11.1)
R14, R15, C30	100k / 10k / 100n	전원 전압 감시	VIN_P / 11 → PB0 ADC (UVLO 8.9 V ~ DAC 최소 10 V 사이 판정)

* 표시 값은 데이터시트로 확인

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DOTECH Co., Ltd.

Sheet: /7. Aux/
File: HMT500(260313A)_aux.kicad_sch

Title: HMT500(260313A) Aux

Size: A3	Date: 2026-09-30	Rev: 0.9
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/8